TEXCYCLE AI: AN AI-POWERED TEXTILE WASTE MARKETPLACE AND CIRCULAR ECONOMY INTELLIGENCE PLATFORM FOR SRI LANKA

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ABSTRACT

Sri Lanka's textile and apparel sector contributes significantly to the national economy, yet production and consumer textile waste present an ongoing circular-economy challenge. Existing digital solutions address identification, sustainability assessment, or marketplace management in isolation. This dissertation details the individual research contribution of Member 04 to TexCycle AI, focusing on the Marketplace Platform and Predictive Analytics layer to evaluate how textile listings are likely to perform in the marketplace.

The component utilizes a structured marketplace dataset of 100,000 synthetic records across 36 variables covering textile characteristics, market competition, and economic indicators. Machine learning models—including XGBoost classifiers and regressors—were implemented for listing success, sales-time estimation, and market trend prediction. The system is deployed via a full-stack architecture combining React/Vite, Node.js/Express, and Python/FastAPI.

Experimental evaluation demonstrates an R^2 of 0.7610 for sales-time prediction, 83.65% accuracy for listing success, 99.54% for demand prediction, and 98.72% for market-trend classification. While these results confirm technical feasibility, real-world deployment accuracy remains unverified due to the synthetic nature of the dataset. It is recommended that future work validate the model using live transactional and inventory data from local apparel manufacturers and industry recyclers.

Keywords: textile waste, artificial intelligence, circular economy, predictive analytics, Sri Lanka.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
F1	F1-score
FASTAPI	Python web framework for APIs
LCA	Life Cycle Assessment
MAE	Mean Absolute Error
ML	Machine Learning
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
XGBoost	Extreme Gradient Boosting
UI	User Interface
UX	User Experience
SLIIT	Sri Lanka Institute of Information Technology
DPP	Digital Product Passport
MVP	Minimum Viable Product
RF	Random Forest

Table 1 List Of Abbreviations

1 INTRODUCTION

1.1 Background and Context

The textile and apparel industry is one of Sri Lanka's major export-oriented sectors. The project literature identifies apparel and textile activity as an important contributor to export earnings and employment, while the generation of production and post-consumer textile waste creates pressure for improved circular resource management [1], [17], [18]. Circular-economy approaches seek to keep materials and products in productive use for longer, reduce dependence on virgin resources and avoid unnecessary disposal [2], [3].

The central challenge addressed by TexCycle AI is not only identifying a piece of textile waste. A usable circular marketplace also needs to estimate material condition, environmental value, commercial value, suitable buyers and likely marketplace performance. The project therefore divides the overall intelligence into four complementary research components. Member 01 determines fabric characteristics and condition; Member 02 generates sustainability intelligence; Member 03 provides traceability, buyer recommendation and AI pricing; and Member 04 provides marketplace prediction.

Member 04 addresses the final predictive question in this chain: after a textile item has been analysed and prepared as a potential listing, how is it likely to perform in the marketplace? This includes estimating whether a listing is likely to become successful, estimating the expected time required for a sale, and classifying the market trend. The component is designed as decision support for sellers rather than as an autonomous decision maker.

1.2 Research Problem

Current textile-waste handling can be fragmented across material assessment, environmental analysis, pricing, buyer identification and transaction management. Even when a seller has sufficient information to publish a textile lot, the seller may still lack evidence about the likely marketplace response. A newly created circular marketplace also faces a cold-start problem because historical transactions are

initially limited. Consequently, manual judgement may dominate listing decisions, inventory planning and seller expectations.

The research problem for Member 04 is therefore defined as follows: how can a marketplace-oriented predictive analytics layer be designed and integrated into a Sri Lankan textile-waste marketplace so that sellers can receive structured predictions about listing success, expected sales time and market trend while clearly communicating the limitations of prototype predictions?

1.3 Research Aim

To design, implement and evaluate a Marketplace Platform and Predictive Analytics component for TexCycle AI that uses structured textile and marketplace features to forecast listing success, expected sales time and market trends, and integrates these predictions into the seller marketplace workflow.

1.4 Research Objectives

1. Design a marketplace prediction dataset schema containing textile, commercial, temporal, geographic and seller-related variables.
2. Prepare and validate a 100,000-record, 36-variable development dataset for predictive marketplace experiments.
3. Implement an XGBoost classifier for listing-success prediction with Sold/Unsold output.
4. Implement an XGBoost regression model for estimating days to sell.
5. Implement market-trend classification for Growing, Stable and Declining states.
6. Integrate prediction services with the Node.js/Express marketplace backend and React/Vite seller interface.
7. Evaluate the predictive models using appropriate classification and regression metrics.
8. Document limitations arising from synthetic data, cold-start conditions, missing real-time market sources and potential model drift.

9. Define a real-data validation plan for future deployment in the Sri Lankan textile and apparel sector.

1.5 Research Questions

1. What marketplace variables are relevant for predicting the performance of textile-waste listings?
2. How effectively can XGBoost-based models estimate listing success and sales time within the development dataset?
3. How can market-trend predictions be integrated into a seller workflow without confusing single-prediction confidence with model-level accuracy?
4. How can the predictive component consume outputs from the other TexCycle AI research components while maintaining clear ownership boundaries?
5. What data and validation requirements are necessary before marketplace predictions can be interpreted as real-world Sri Lankan evidence?

1.6 Scope of the Individual Contribution

The individual scope covers the marketplace predictive dataset, prediction services for listing success, sales time and market trend, marketplace intelligence presentation, prediction history and integration of these outputs into the seller workflow. Pricing and buyer recommendation are treated as upstream inputs from Member 03 rather than as Member 04 research outputs. Fabric classification and sustainability modelling are likewise treated as upstream components whose outputs are consumed by the marketplace predictor.

1.7 Significance of the Research

The component adds a forward-looking layer to the marketplace. A listing can contain correct textile information and a reasonable price yet still require an estimate of how quickly it may sell or whether its category is growing or declining. Predictive analytics can support seller planning, inventory prioritisation and marketplace monitoring. The research significance is strongest when the forecasts are interpreted

as decision-support outputs and when uncertainty, data quality and model limitations remain visible.

1.8 Chapter Structure

Chapter 1 introduces the problem, objectives and scope. Chapter 2 reviews circular textile research, AI marketplace prediction and related digital systems. Chapter 3 presents the research methodology and dataset strategy. Chapter 4 defines the product and solution design. Chapter 5 documents implementation, integration and testing. Chapter 6 presents model results and validation. Chapter 7 discusses findings, limitations and responsible interpretation. Chapter 8 addresses commercialization and sustainability. Chapter 9 concludes the research and defines future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Circular Economy and Textile Waste

The circular economy literature emphasizes extending product and material utility through reuse, repair, remanufacturing and recycling rather than relying on a linear take-make-dispose model [2], [3]. Textile waste presents a particular challenge because fibre composition, blends, contamination, condition and local collection systems affect whether a material can be recovered economically. Systematic reviews identify technological, economic, stakeholder and supply-chain barriers in textile-waste management [4], while environmental studies highlight the importance of consistent assessment methods [5], [23].

For a digital textile marketplace, circularity is not achieved simply by placing waste on a website. The platform must support a chain of decisions that converts an uncertain waste item into structured inventory. The TexCycle AI architecture addresses this by combining material intelligence, sustainability intelligence, commercial intelligence and predictive marketplace analytics.

2.2 Artificial Intelligence for Textile Sorting and Assessment

Computer vision and machine learning have increasingly been explored for textile and fibre recognition [6], [8], [20], [22], [24], [25]. CNN-based methods can classify visual material patterns, while spectroscopy-based approaches can provide additional information for fibre identification. However, research also highlights the challenge of blended fibres and the dependence of image-based systems on dataset quality and acquisition conditions.

The relevance to Member 04 is indirect but important. Marketplace prediction should not operate on arbitrary or unstructured inputs. It consumes structured information produced earlier in the pipeline, such as fabric type, condition, health score, repairability and remaining lifespan. This creates a dependency on upstream data quality and reinforces the need to preserve model boundaries.

2.3 Sustainability Intelligence and Marketplace Decision Making

Textile circularity decisions should account for environmental value as well as commercial value [3], [5], [6], [11], [23]. The Member 02 sustainability component produces structured indicators that can be consumed by downstream marketplace intelligence. This enables a marketplace record to contain both environmental and commercial attributes instead of treating sustainability as an unrelated report.

For Member 04, sustainability values are therefore input features rather than targets. The predictive question is not whether a textile is environmentally sustainable, but how the marketplace may respond to a listing whose sustainability and material characteristics are already known.

2.4 Machine Learning for Structured Marketplace Prediction

Structured marketplace datasets contain mixtures of categorical and numerical features. Gradient-boosted decision-tree methods are useful for such data because they can model nonlinear relationships and interactions without requiring image-based representations. XGBoost is a scalable gradient-boosting framework widely used for structured prediction tasks [26].

Three prediction tasks are particularly relevant. Listing-success prediction is a binary classification problem, where the target represents whether a listing reaches a sold outcome. Sales-time prediction is a regression problem, where the target is the number of days to sell. Market-trend prediction is a multiclass classification problem with categories such as Growing, Stable and Declining.

2.5 Marketplace Recommendation and Commercial Intelligence

Recommendation systems rank candidate items or users according to compatibility. Within TexCycle AI, buyer recommendation is an upstream Member 03 contribution. Member 04 does not replace that recommendation. Instead, buyer type and predicted price can be used as marketplace features when estimating listing performance. This separation allows the recommendation and prediction tasks to be evaluated independently.

2.6 Digital Traceability and Marketplace Records

Digital traceability frameworks seek to maintain product information across stages of a lifecycle [7], [8], [12]. In a circular textile marketplace, a prediction record should be associated with a listing and a model version so that later users can understand which prediction was produced at which point in the lifecycle. This is especially important when market conditions change.

2.7 Research Gap

Existing direction	**Primary focus**	**Remaining limitation addressed by Member 04**
Textile classification	Material/fibre identification	Does not forecast marketplace behaviour
Sustainability assessment	Environmental impacts	Often separated from listing performance
Digital Product Passports	Traceability and transparency	Do not inherently predict sales behaviour
Price prediction	Commercial valuation	Does not necessarily estimate listing outcome or sales time
Recommendation systems	Matching users/items	Do not necessarily forecast listing performance
Marketplace prediction	Demand/sales behaviour	Rarely integrated with textile condition and circularity data

Table 2 Research directions and the Member 04 gap

2.8 Conceptual Research Position

The research gap is therefore not that marketplace prediction or textile marketplaces have never been studied. The stronger claim is that TexCycle AI integrates marketplace prediction with textile intelligence, sustainability intelligence, commercial intelligence and lifecycle traceability in a Sri Lankan-focused textile-waste context. The Member 04 contribution provides the predictive layer that estimates likely marketplace behaviour after the item has been transformed into structured circular inventory.

3 RESEARCH METHODOLOGY

3.1 Research Approach

The research follows an applied Design Science and experimental machine-learning approach. Design Science is appropriate because the study produces an artefact—a marketplace predictive component—and evaluates whether the artefact can solve a defined decision-support problem. The experimental component evaluates predictive models using held-out development data and standard performance measures.

3.2 Research Process

1. Problem definition and scope separation
2. Literature and domain analysis
3. Marketplace feature design
4. Synthetic dataset generation and quality checks
5. Exploratory analysis and preprocessing
6. Model selection and training
7. Model evaluation
8. API integration
9. Frontend integration
10. Functional and failure-path testing
11. Interpretation of results and limitations
12. Real-data validation planning

3.3 Dataset Strategy

The current Member 04 marketplace dataset contains 100,000 records and 36 variables. The source documentation explicitly identifies this dataset as synthetic/prototype data. It was created to validate the complete machine-learning and application workflow while real marketplace data are still being collected.

Consequently, the dataset is suitable for pipeline experimentation but should not be treated as a representative sample of the Sri Lankan textile market.

Group	Representative variables	Role
Textile	fabric type, condition, health score, defect area, repairability, remaining lifespan	Material context
Quantity	weight_kg	Lot scale
Sustainability	sustainability score, green score, CO2 saving, water saving	Circular context
Commercial	predicted price, buyer type, demand level	Marketplace context
Temporal	listing month, week, day, hour, season	Time effects
Geographic	district, province	Regional context
Market	competition, competitor count, export demand, economic index, market growth, average market price	Market conditions
Seller	seller rating, seller experience	Seller context
Engagement	weekly views, wishlist count, click rate	Marketplace engagement
Targets	listing status, days to sell, market trend	Prediction targets

Table 3 Member 04 marketplace dataset structure

3.4 Target Variables

Prediction task	Target	Problem type	Primary model
Listing success	listing_status (Sold/Unsold)	Binary classification	XGBoost Classifier
Sales time	days_to_sell	Regression	XGBoost Regressor
Market trend	market_trend (Growing/Stable/Declining)	Multiclass classification	Trained classification model + label encoder
Demand level	demand level (Low/Medium/High)	Multiclass classification	Trained classifier

Table 4 Marketplace predictive targets

3.5 Data Preparation

Data preparation follows the sequence data gathering, cleaning, exploratory analysis, feature engineering, train/test splitting, preprocessing, baseline modelling, training, tuning, evaluation and model persistence. Categorical variables require consistent encoding between training and inference. Numerical variables must be validated for missing, invalid or extreme values. Temporal fields should be represented consistently so that listing-time effects can be learned without creating incompatible inference features.

3.6 Feature Engineering

Feature engineering combines the structured outputs of the upstream intelligence layers with marketplace context. Fabric type and condition describe the material. Health, repairability and remaining lifespan provide condition signals. Sustainability and green scores describe environmental characteristics. Predicted price and buyer type provide commercial context. Competition, competitor count, market growth and average market price describe the market environment. Engagement variables such as weekly views, wishlist count and click rate provide behavioural signals where available.

An important implementation principle is that the prediction layer should not silently manufacture highly precise values when a real data source is absent. The marketplace controller documents neutral defaults for certain fields, including export demand, economic index and seller rating, where a real source was not available in the current prototype. This design makes the limitation explicit rather than presenting invented values as observations.

3.7 Model Selection Rationale

XGBoost was selected for the listing-success and sales-time tasks because the inputs are structured tabular variables with nonlinear interactions. The trend model uses a trained classifier with a persisted label encoder so that category mappings remain consistent between training and inference. The project configuration also contains

explicit model paths for marketplace trend, listing success, sales time and demand prediction artefacts.

3.8 Evaluation Metrics

Task	Metrics	Interpretation
Binary listing success	Accuracy, precision, recall, F1, confusion matrix	Ability to distinguish Sold and Unsold
Sales time	MAE, RMSE, R^2	Absolute error, squared error and explained variance
Market trend	Accuracy, precision, recall, F1, confusion matrix	Correct classification across trend classes
Demand level	Accuracy, precision, recall, F1	Correct classification across demand levels

Table 5 Evaluation framework

3.9 Responsible Evaluation Principle

The methodology separates model-level evaluation metrics from individual prediction confidence. For example, a single prediction may display a high probability for one trend class, but that probability is not equivalent to overall validation accuracy. This distinction is essential for responsible reporting [9].

3.10 Reproducibility

Reproducibility requires recording the dataset version, feature list, target list, train/test split, random seed where applicable, preprocessing configuration, model algorithm and hyperparameters, evaluation metrics, model file path and API version. Model version and prediction timestamp should also be persisted with marketplace prediction records.

4 PRODUCT AND SOLUTION DESIGN

4.1 Functional Requirements

ID	Requirement	Priority	Verification
FR-01	Accept structured marketplace prediction inputs	High	API test
FR-02	Predict listing success probability/class	High	Model inference test
FR-03	Estimate expected sales time	High	Regression test
FR-04	Classify market trend	High	Trend inference test
FR-05	Provide demand intelligence where model data are available	Medium	Prediction test
FR-06	Persist prediction history with listing/model context	High	Database test
FR-07	Display predictions in seller marketplace workflow	High	UI integration test
FR-08	Handle unavailable real market inputs transparently	High	Fallback test
FR-09	Support model version traceability	Medium	Audit test
NFR-01	Maintain consistent training/inference preprocessing	High	Artifact verification
NFR-02	Avoid misleading precision in missing-data fields	High	Edge-case test
NFR-03	Provide responsive marketplace intelligence UI	Medium	Frontend test

Table 6 Functional and non-functional requirements

4.2 System Architecture

The Member 04 solution sits inside a layered architecture. React/Vite provides the presentation layer. Node.js/Express manages marketplace application services, authentication, listings, offers, transactions and orchestration. Python/FastAPI hosts

machine-learning services. Database storage persists marketplace records and prediction histories. The predictive component consumes structured outputs from upstream research components and returns marketplace intelligence to the seller workflow.

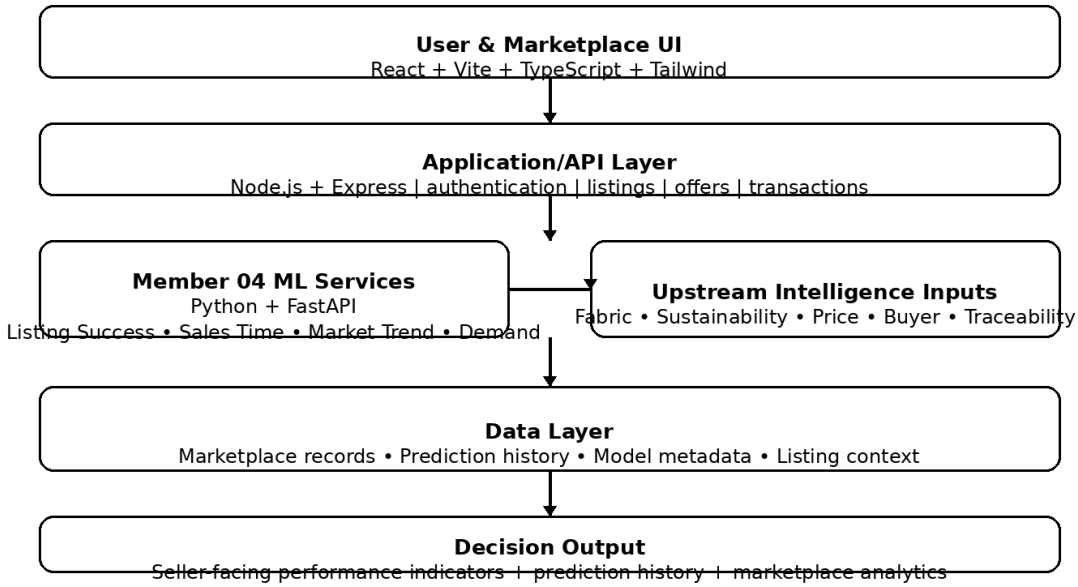


Figure 4.1: Member 04 Marketplace Platform and Predictive Analytics architecture

4.3 Prediction Data Flow

The prediction flow begins after the seller has completed or received the upstream textile analysis. Relevant structured values are assembled into the marketplace prediction payload. The backend validates the payload and passes it to the ML service. The selected models generate listing-success, sales-time and trend outputs. The results are stored against the marketplace record and returned to the seller interface.

4.4 Seller Workflow Design

1. Upload textile waste and basic lot information
2. Run fabric and condition analysis
3. Receive sustainability intelligence

4. Receive reference price and buyer recommendation
5. Run marketplace predictive analytics
6. Review listing success, sales-time and market-trend outputs
7. Adjust listing details if required
8. Publish marketplace listing
9. Monitor offers and transactions
10. Retain prediction history for later comparison

4.5 Marketplace Prediction Record

Field group	Examples
Identity	prediction_id, listing_id, seller_id
Input context	fabric_type, condition, weight_kg, health_score
Commercial context	predicted_price, buyer_type, demand_level
Market context	competition, competitor_count, market_growth, average_market_price
Engagement context	weekly_views, wishlist_count, click_rate
Predictions	listing_success, expected_days_to_sell, market_trend
Audit metadata	model_version, timestamp, prediction source

Table 7 Suggested marketplace prediction record

4.6 Separation of Member Responsibilities

The architecture deliberately maintains research boundaries. Member 03 owns AI pricing, buyer recommendation and traceability. Member 04 consumes those outputs and focuses on marketplace performance prediction. This prevents duplication and makes it possible to evaluate the contribution of each research component independently.

4.7 User Roles

Role	Member 04 relevant capability
Seller	Run predictions, review listing performance indicators, create/manage listings

Buyer	View marketplace listings and relevant prediction-supported information where exposed
Admin	Monitor marketplace analytics and prediction operation
Super Admin	Manage model versions, configuration and global analytics

Table 8 Role-based marketplace responsibilities

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Technology Stack

Layer	Technology	Member 04 role
Frontend	React, Vite, TypeScript, Tailwind CSS, React Router, Lucide	Marketplace intelligence UI
Backend	Node.js, Express	Listings, marketplace operations, prediction orchestration
ML service	Python, FastAPI	Model inference endpoints
ML algorithms	XGBoost and trained classifiers	Prediction
Persistence	Database-backed storage / MongoDB project architecture	Listings and prediction history
Model artifacts	Pickle/joblib artefacts and label encoders as configured	Persisted model inference

Table 9 Member 04 implementation technology stack

5.2 Model Artefacts and Configuration

The project configuration explicitly defines separate model paths for marketplace intelligence. The price and demand artefacts are grouped with marketplace commercial intelligence, while marketplace trend, listing-success and sales-time models have dedicated paths. Member 04 ownership is centred on the latter predictive analytics layer. The configured artefacts include marketplace_trend_model.pkl, marketplace_trend_label_encoder.pkl, listing_success_model.pkl, listing_success_encoder.pkl and sales_time_prediction_model.pkl.

5.3 Listing Success Prediction

Listing success is treated as a binary classification task with listing_status as the target and Sold/Unsold as the output states. The XGBoost classifier receives structured marketplace features and returns a predicted outcome and, where supported by the model interface, a probability distribution. The UI should describe the probability as a model output rather than a guarantee that a listing will sell.

5.4 Sales-Time Prediction

Sales-time prediction estimates the number of days required for a listing to sell. The current approach uses an XGBoost regressor. The output can support seller planning, inventory prioritisation and marketplace analytics. Because a new marketplace has limited historical transactions, the prediction is sensitive to the quality and representativeness of the training data.

5.5 Market Trend Prediction

Market trend prediction classifies the market state into Growing, Stable or Declining. A persisted label encoder maintains the mapping between model output and display labels. The implementation must distinguish a single prediction probability from the model's validation accuracy. A prediction such as "Stable: 99.72%" is an instance-level confidence/distribution and not evidence that the model is 99.72% accurate overall.

5.6 Demand Intelligence

Demand level is included in the marketplace dataset and is available as a predictive marketplace signal. The research draft reports 99.54% demand prediction accuracy for the development evaluation. This result is retained as a reported prototype metric and must be interpreted with the same synthetic-data limitation as the other marketplace models.

5.7 Backend Integration

The marketplace controller invokes the ML marketplace analysis utility and combines model results with real inputs when they are available. The implementation explicitly defines neutral defaults for data sources that do not yet exist in the prototype, including export demand, economic index and seller rating. This is preferable to silently inserting arbitrary market observations. The controller also contains a deterministic fallback path so that the marketplace can remain operational when the trained prediction service is unavailable.

5.8 Frontend Integration

The seller-facing interface provides dashboard, upload, listing and analysis-history workflows. Marketplace prediction outputs are presented after the relevant upstream analysis is available. The UI can expose demand, trend, listing-success and sales-time information together because these indicators answer different questions: current demand, direction of the market, likelihood of conversion and expected time to conversion.

5.9 Prediction History and Auditability

Prediction history is important because marketplace conditions change. A historical prediction should retain the listing context and model metadata used at the time of inference. This allows future analysis to compare predictions with actual listing outcomes and supports model monitoring.

5.10 Testing Strategy

Test ID	Scenario	Expected result
M4-T01	Valid marketplace payload	Prediction service returns structured output
M4-T02	Missing optional market context	Neutral documented defaults are used
M4-T03	Invalid numerical input	Validation rejects or safely normalizes the request
M4-T04	Listing-success inference	Sold/Unsold output is returned
M4-T05	Sales-time inference	Non-negative days-to-sell estimate is returned
M4-T06	Trend inference	Growing/Stable/Declining label is returned
M4-T07	Model unavailable	Controlled fallback/error response is produced
M4-T08	Prediction history	Prediction can be retrieved with metadata
M4-T09	Frontend rendering	Prediction cards render without layout overflow
M4-T10	Production build	Frontend builds without compilation errors

Table 10 Representative Member 04 verification tests

5.11 Edge Cases

- Unknown fabric type or category
- Very small or very large textile weight
- Missing seller rating
- Missing macroeconomic indicator
- No historical views or wishlist interactions
- New seller with minimal experience
- New listing with no historical marketplace data
- Unavailable ML service
- Malformed model response
- Model-version mismatch

5.12 Deployment Considerations

The component is suitable for a separated service deployment because the frontend, application backend and ML service communicate through defined interfaces. Production deployment should add authentication for prediction endpoints, monitoring, model-version control, structured logging, secure environment variables and health checks. Real-time external market feeds should only be introduced after data provenance and licensing are addressed.

6 RESULTS AND VALIDATION

6.1 Development Dataset

The current Member 04 development dataset contains 100,000 synthetic records and 36 variables. The dataset supports experimentation across textile characteristics, commercial context, market context, seller characteristics, engagement and prediction targets. Because the records are synthetic, the resulting evaluation measures indicate performance on the generated relationships rather than validated real-world market behaviour.

6.2 Model Evaluation Results

Model	Metric	Reported result	Interpretation
Sales-time prediction	R^2	0.7610	Moderate explained variance on the development evaluation
Demand prediction	Accuracy	99.54%	High development-set classification accuracy; requires real-data validation
Listing-success prediction	Accuracy	83.65%	Development-set classification performance
Market-trend prediction	Accuracy	98.72%	Development-set classification performance

Table 11 Member 04 marketplace predictive evaluation results

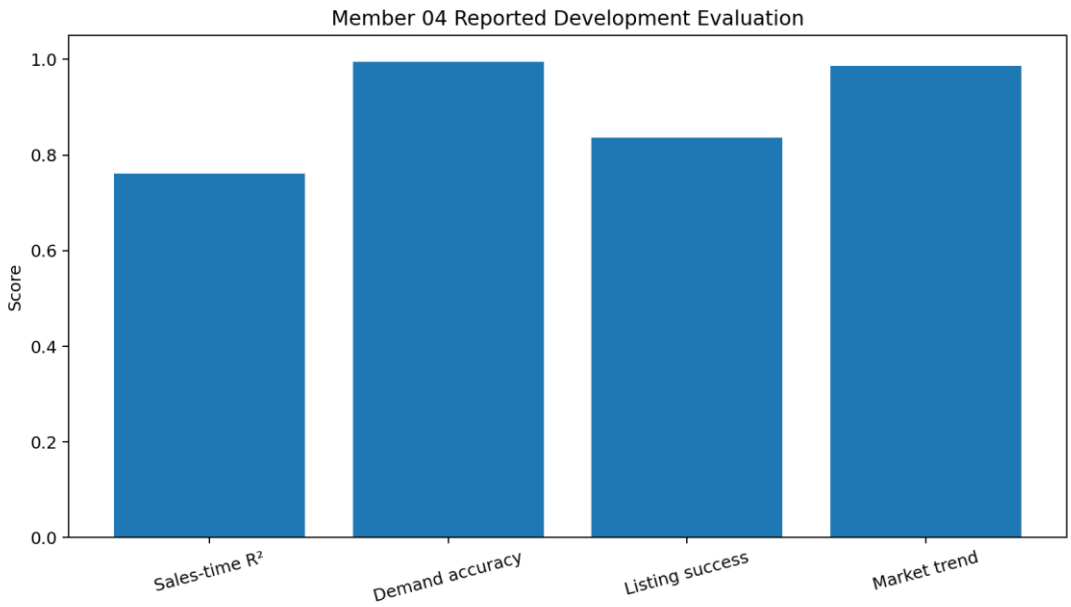


Figure 6.1: Reported development evaluation for Member 04 predictive models

6.3 Interpretation of Listing-Success Accuracy

The reported listing-success accuracy of 83.65% indicates that the classifier correctly classified a large majority of development-set cases. Accuracy alone, however, is insufficient to determine whether the model is equally reliable for Sold and Unsold cases. A complete final validation should report class balance, precision, recall, F1-score and the confusion matrix. In particular, an imbalanced dataset could make accuracy appear stronger than the model's performance on the minority class.

6.4 Interpretation of Sales-Time R^2

The reported sales-time R^2 of 0.7610 indicates that approximately 76.10% of the variance in the development target is explained by the fitted model under the evaluation conditions. R^2 does not directly state the typical number of days by which a prediction is wrong. MAE and RMSE should therefore be reported alongside R^2 in the final experimental run. The project methodology explicitly identifies MAE, RMSE and R^2 as the appropriate regression measures.

6.5 Interpretation of Demand Accuracy

The reported 99.54% demand prediction accuracy is high and should be interpreted cautiously because the underlying dataset is synthetic. A generated dataset can contain deterministic or highly learnable relationships that do not exist in a live market. The value is therefore useful as evidence that the implemented training and inference pipeline works on the development data, but not as evidence that demand can be predicted with 99.54% accuracy in Sri Lanka.

6.6 Interpretation of Market-Trend Accuracy

The reported 98.72% market-trend accuracy demonstrates strong development-set classification performance. However, market trend is inherently time-dependent and vulnerable to distribution shift. A model trained on synthetic relationships or historical assumptions may not capture sudden changes in consumer behaviour, export demand, competitor supply, economic conditions or seasonal effects. Temporal validation and future-period testing are therefore required.

6.7 Integrated Demonstration Scenario

The project draft describes a 50-kilogram Denim Textile Waste example in which upstream intelligence produces a reference price of LKR 260.15/kg, a listed price of LKR 109.16/kg and an estimated total value of LKR 13,007.50. The marketplace predictive layer then presents high demand, a Growing market trend with 70% confidence, a 74% listing-success probability and an estimated sales time of approximately eighteen days. These values are retained as prototype interface outputs, not independent measurements of the Sri Lankan textile market.

6.8 Validation of End-to-End Workflow

Validation level	Evidence	Conclusion
Model level	Reported predictive metrics	Pipeline produces measurable model outputs
Service level	Structured prediction API and fallback design	Application can consume prediction results
Integration level	Seller workflow combines upstream and marketplace outputs	Predictive layer is connected to marketplace operation

User level	Seller-facing cards/history and role workflow	Decision-support presentation is feasible
Real-market level	No sufficiently validated real transaction dataset yet	Real-world predictive accuracy remains unproven

Table 12 Evidence boundaries for Member 04 validation

6.9 Hypothesis Assessment

H1: A structured marketplace prediction layer can be implemented within the TexCycle AI architecture. Supported by the completed architecture, prediction models, API integration and workflow design.

H2: The current models demonstrate useful predictive performance on the development dataset. Partially supported by the reported metrics, with the explicit limitation that the dataset is synthetic.

H3: The models are ready to represent real Sri Lankan marketplace accuracy. Not supported at the present stage because real transaction and longitudinal marketplace validation remain limited.

7 DISCUSSION

7.1 Main Findings

The main finding is that marketplace predictive analytics can be implemented as a distinct intelligence layer after textile, sustainability and commercial attributes have been structured. The component does not need to replace seller judgement. Instead, it provides additional evidence that can be considered before publication and while monitoring a listing.

7.2 Integrated Intelligence Value

The four research components answer complementary questions. Fabric Intelligence asks what the textile is and what condition it is in. Sustainability Intelligence asks what environmental value it represents. Commercial Intelligence asks what it may be worth and who may buy it. Marketplace Predictive Analytics asks how it is likely to perform. The combination creates a progression from unstructured textile waste to structured AI-assessed material, sustainable commercial inventory, predictive marketplace listing and traceable circular transaction.

7.3 Cold-Start Problem

A major limitation for marketplace prediction is the cold-start problem. At the beginning of a marketplace, few historical transactions exist. Variables such as views, click rate, wishlist count and seller rating may be sparse. The current design therefore allows the model to consume broader material and market features, while missing values must be handled explicitly. Real deployment should progressively replace synthetic assumptions with observed marketplace data.

7.4 Model Drift and Market Volatility

Textile demand can change due to fashion cycles, seasonal production, export demand, commodity prices, policy changes and competitor behaviour. A model that performs well at one time may degrade later. Continuous evaluation, temporal hold-out tests and drift monitoring are therefore essential. The model registry should

record active versions and allow historical predictions to remain associated with the version that produced them.

7.5 Explainability and Trust

Marketplace users may interpret AI outputs as promises if the interface is overly confident. The design should therefore use language such as “estimated sales time,” “predicted listing success,” and “market trend classification.” It should also distinguish confidence from accuracy. This is especially important for a marketplace where predictions can influence seller pricing, inventory and expectations.

7.6 Ethical Considerations

Predictive marketplace systems can create unfair outcomes if historical data reflect unequal visibility, seller reputation or buyer access. A high seller rating can become self-reinforcing if it increases predicted success and therefore visibility. The platform should monitor performance across seller groups and avoid using protected or irrelevant personal attributes. Prediction outputs should remain decision-support indicators rather than automatic exclusion rules.

7.7 Data Privacy and Security

Marketplace records may contain seller identities, transaction information, pricing and business relationships. Prediction services should receive only the fields required for inference, use authenticated endpoints and store model outputs with appropriate access controls. Sensitive information should not be exposed through public prediction APIs. Audit logs should identify model version and prediction time without unnecessarily duplicating confidential data.

7.8 Research Contribution

The individual contribution is the implementation and evaluation of a predictive marketplace layer that uses a broad structured dataset and connects three marketplace prediction tasks to a functioning circular textile marketplace. The contribution is not the invention of XGBoost or marketplace prediction itself. Its research value lies in

the integration, textile-specific feature design, Sri Lankan localisation, workflow integration and explicit validation boundaries.

7.9 Comparison with Existing Approaches

Dimension	Conventional marketplace	TexCycle AI Member 04
Material context	Often generic product attributes	Textile type, condition and sustainability context
Circularity context	Usually absent	Upstream sustainability outputs can be consumed
Listing outcome	Manual judgement or simple analytics	Predictive listing-success model
Sales time	Observed retrospectively	Estimated before/while listing
Trend	Dashboard statistics	Trend classification model
Traceability	Platform-dependent	Connected to broader textile lifecycle
Localisation	Generic markets	Sri Lankan districts/provinces and textile context

Table 13 Conceptual comparison of marketplace intelligence

7.10 Limitations

- Synthetic 100,000-record development dataset limits real-world validity.
- Real transaction history and sale-time observations remain limited.
- Some market-context fields currently use neutral defaults when a real source is unavailable.
- Cold-start conditions reduce the availability of behavioural features.
- Market conditions can drift over time.
- The reported evaluation set does not by itself establish fairness across seller groups.
- The final deployment should validate classification metrics beyond accuracy and regression metrics beyond R^2.

7.11 Responsible Claim Boundary

The strongest defensible claim is that the Member 04 prototype demonstrates technical feasibility of integrated marketplace prediction on a structured development dataset. It is not defensible to claim that the current metrics prove 83.65% real-market listing-success accuracy, 98.72% real-world trend accuracy or 99.54% real-world demand accuracy in Sri Lanka. These values require future validation against real, time-stamped marketplace observations.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Target Users

The primary users of the predictive marketplace layer are textile-waste sellers, apparel manufacturers, recyclers, designers and other organisations that publish reusable or recyclable textile lots. Buyers benefit indirectly because improved listing quality and marketplace discovery can increase the availability of suitable materials.

8.2 Value Proposition

The predictive component adds value by reducing uncertainty before a listing is published. A seller can review the estimated listing outcome, sales-time expectation and market trend alongside price, sustainability and buyer information. This can support more informed listing decisions, inventory planning and follow-up activities.

8.3 Revenue Opportunities

The predictive layer can support several commercialization models without forcing predictions to become mandatory. Basic marketplace access can remain free or low-cost, while premium analytics can provide advanced market insights, historical performance reports, trend dashboards and seller performance benchmarking. Enterprise users could access API-based analytics subject to appropriate data governance.

8.4 Sustainability Contribution

A marketplace prediction layer contributes to sustainability indirectly by improving the chance that textile waste is successfully matched with a useful downstream application. Faster and more reliable discovery can reduce the time materials remain unused. However, the predictive model itself should not be described as measuring environmental impact. Environmental impact indicators remain within the sustainability intelligence component.

8.5 Sri Lankan Market Context

The project literature identifies Sri Lanka's apparel and textile sector as economically important and emphasises the need for sustainable and transparent industry practices [1], [17], [18]. A locally oriented marketplace can incorporate districts, provinces, buyer categories, local currency and sector-specific market characteristics. Future commercialisation should be validated with apparel factories, recyclers, exporters, designers and waste aggregators.

8.6 Stakeholder Benefits

Stakeholder	Potential benefit
Textile seller	Better listing decisions and expected sales-time information
Recycler	Improved visibility into supply and market trends
Designer/artisan	Faster discovery of suitable textile materials
Marketplace operator	Analytics for inventory and platform performance
Buyer	More informative listings and marketplace discovery
Research/industry partner	Structured evidence for future circular-market studies

Table 14 Stakeholder benefits

8.7 Sustainable Growth Strategy

1. Start with pilot suppliers and recyclers
2. Collect verified real transaction and listing data
3. Validate model predictions against actual outcomes
4. Retrain models using representative Sri Lankan observations
5. Introduce temporal drift monitoring
6. Expand regional coverage after local validation
7. Add advanced analytics only after data quality is sufficient

8.8 Commercial Risk

Over-reliance on predictions could create commercial disputes if users interpret estimates as guarantees. The platform should therefore communicate that predicted listing success, sales time and trends are advisory. The actual market outcome depends on buyer behaviour, price, material quality, logistics and external economic conditions.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This dissertation presented the Member 04 Marketplace Platform and Predictive Analytics component of TexCycle AI. The component was designed to answer a specific question within the wider circular textile workflow: how is an analysed textile listing likely to perform in the marketplace? The solution combines a 100,000-record, 36-variable synthetic development dataset with predictive models for listing success, sales time and market trend, and it provides demand-related intelligence within the marketplace analytics layer.

The implementation uses XGBoost for listing-success classification and sales-time regression and a trained classification model with label encoding for market trend. The reported development evaluation includes sales-time R^2 of 0.7610, demand accuracy of 99.54%, listing-success accuracy of 83.65% and market-trend accuracy of 98.72%. These results demonstrate the feasibility of the implemented predictive pipeline under the current development conditions.

The most important qualification is that the marketplace dataset is synthetic. The results therefore cannot be presented as validated Sri Lankan market accuracy. The research instead establishes an operational foundation for future validation using real listings, prices, views, offers, transactions, sale dates and market conditions. This distinction is central to the scientific integrity of the component.

9.2 Research Contributions

1. A structured 36-variable marketplace prediction dataset design
2. An integrated listing-success prediction component
3. An integrated sales-time regression component
4. A market-trend classification component with persistent label mapping
5. Demand-related marketplace intelligence
6. Backend and frontend integration into the seller workflow

7. Prediction history and model-version traceability concepts
8. A responsible evaluation boundary separating prototype performance from real-market claims

9.3 Future Work

1. Collect verified real Sri Lankan textile marketplace data.
2. Record actual listing and sale dates to improve sales-time validation.
3. Introduce temporal cross-validation and future-period evaluation.
4. Report precision, recall, F1 and confusion matrices for classification tasks.
5. Report MAE and RMSE alongside R^2 for sales-time regression.
6. Monitor model drift and retrain using newly verified observations.
7. Replace neutral market defaults with licensed, auditable data sources.
8. Develop explainability views for seller-facing predictions.
9. Evaluate fairness across seller and geographic groups.
10. Introduce dynamic pricing only after robust real-data validation.
11. Expand demand and trend modelling with longitudinal market data.
12. Integrate verified external economic and export-demand indicators.

9.4 Final Research Statement

The Member 04 contribution demonstrates that predictive marketplace intelligence can be integrated into a circular textile marketplace without replacing the specialist responsibilities of the other research components. The research provides a technically coherent prototype and a clear pathway from synthetic development data to real-world validation. The next research milestone is not to increase the apparent accuracy of the current synthetic models, but to obtain representative Sri Lankan observations and test whether the same architecture continues to perform under genuine marketplace conditions.

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GLOSSARY

Term	Definition
Listing success	A classification output indicating whether a marketplace listing is expected to reach a sold outcome under the model definition.
Sales time	Estimated number of days required for a listing to reach the defined sale outcome.
Market trend	Classification of marketplace direction into categories such as Growing, Stable or Declining.
Demand level	A categorical representation of marketplace demand, such as Low, Medium or High.
Synthetic dataset	Data generated according to predefined rules for development or pipeline testing rather than collected from real marketplace observations.
Cold start	A condition in which a new marketplace has limited historical interaction and transaction data.
Model drift	A reduction in model reliability caused by changes in data distributions or relationships over time.
Prediction history	Persisted records of prior model outputs together with listing and model context.
XGBoost	A scalable gradient-boosted decision-tree framework used for structured machine-learning prediction.
Marketplace intelligence	Structured analytics and predictions used to support marketplace decisions.
Feature engineering	Transformation and construction of input variables so they can be used consistently by a machine-learning model.
Label encoder	A fitted mapping between categorical class labels and model-compatible numeric representations.

Table 15 Glossary

APPENDICES

Appendix A: Member 04 Model Specification

Model	Target	Type	Algorithm	Reported metric
Listing Success	listing_status	Binary classification	XGBoost Classifier	Accuracy 83.65%
Sales Time	days_to_sell	Regression	XGBoost Regressor	R^2 0.7610
Demand	demand level	Multiclass classification	Trained classifier	Accuracy 99.54%
Market Trend	market_trend	Multiclass classification	Trained classifier + encoder	Accuracy 98.72%

Table 16 Model Specification

Appendix B: Marketplace Feature Dictionary

Feature	Type	Meaning	Role
fabric_type	Categorical	Textile material category	Input
condition	Categorical	Observed/assessed textile condition	Input
health_score	Numeric	Condition score from upstream analysis	Input
defect_area	Numeric	Defect-related condition indicator	Input
repairability	Categorical	Repairability assessment	Input
remaining_lifespan	Numeric	Estimated remaining useful life	Input
weight_kg	Numeric	Textile lot weight	Input
sustainability_score	Numeric	Upstream sustainability indicator	Input
green_score	Numeric	Upstream green indicator	Input
co2_saving	Numeric	Upstream CO2-saving indicator	Input
water_saving	Numeric	Upstream water-saving indicator	Input
predicted_price	Numeric	Upstream commercial price prediction	Input

buyer_type	Categorical	Upstream buyer category	Input
demand_level	Categorical	Marketplace demand signal	Input/derived
listing_month	Categorical/Numeric	Listing month	Input
listing_week	Numeric	Week of year	Input
listing_day	Numeric	Day-related listing feature	Input
listing_hour	Numeric	Hour of listing	Input
season	Categorical	Seasonal context	Input
district	Categorical	Sri Lankan district	Input
province	Categorical	Sri Lankan province	Input
market_competition	Numeric	Competition indicator	Input
competitor_count	Numeric	Number of competitors	Input
export_demand	Numeric	Export-demand indicator/default	Input
economic_index	Numeric	Economic context/default	Input
market_growth	Numeric	Market growth indicator	Input
average_market_price	Numeric	Average market price	Input
seller_rating	Numeric	Seller rating/default	Input
weekly_views	Numeric	Listing views	Input
wishlist_count	Numeric	Wishlist interactions	Input
click_rate	Numeric	Listing click rate	Input
seller_experience	Numeric	Seller experience indicator	Input
listing_status	Categorical	Sold/Unsold target	Target
days_to_sell	Numeric	Sale-time target	Target
market_trend	Categorical	Trend target	Target

Table 17 Marketplace feature dictionary

Appendix C: Model Inference Pseudocode

RECEIVE marketplace prediction request

VALIDATE required and optional fields

NORMALIZE categorical and numeric inputs

APPLY documented defaults only where a real source is unavailable

LOAD active model artifacts

GENERATE listing-success prediction

GENERATE sales-time estimate

GENERATE market-trend classification

GENERATE demand output where enabled

ATTACH model version and timestamp

PERSIST prediction history

RETURN structured marketplace intelligence

Appendix D: Test Catalogue

Test	Input condition	Expected outcome
D-01	Complete valid payload	All enabled predictions returned
D-02	Missing export demand	Documented neutral default applied
D-03	Missing seller rating	Documented neutral default applied
D-04	Invalid weight	Validation/failure response
D-05	Missing model artifact	Controlled fallback/error
D-06	Trend label mapping	Readable category returned
D-07	History retrieval	Correct listing/model context
D-08	Frontend narrow viewport	No horizontal overflow
D-09	API timeout	User-safe error state
D-10	Model version changed	New version recorded in history

Table 18 Test Catalogue

Appendix E: Evidence Capture Plan

The final thesis should use only screenshots captured from the tested implementation. No synthetic screenshots are fabricated in this dissertation. Recommended evidence includes: marketplace prediction input/result, listing-success output, sales-time output, market-trend output, prediction history, seller listing workflow and model/version information.

Evidence	Supports	Recommended chapter
E1	Prediction input and validation	4, 5
E2	Listing-success output	5, 6
E3	Sales-time output	5, 6
E4	Trend output	5, 6
E5	Prediction history	5, 6
E6	Seller marketplace workflow	4, 5, 6
E7	Model/version evidence	5, 7

Table 19 Evidence Mapping

Appendix F: Reproducibility Checklist

- Record dataset file/version.
- Record synthetic-data generation rules.
- Record train/test split and random seed.
- Record exact feature and target lists.
- Record preprocessing configuration.
- Record model and library versions.
- Record evaluation metrics.
- Record model artifact filenames.
- Record active model version.
- Record API contract.
- Run tests after model changes.
- Capture final evidence screenshots from the tested system.

Appendix G: Member 04 Contribution Timeline

Period	Contribution
Week 1	Research problem review and marketplace scope separation.
Week 2	Literature review on marketplace analytics and circular textile systems.

Week 3	Defined marketplace feature and target variables.
Week 4	Prepared synthetic dataset strategy and quality checks.
Week 5	Explored feature distributions and target relationships.
Week 6	Prepared preprocessing and baseline prediction workflow.
Week 7	Trained listing-success classifier.
Week 8	Trained sales-time regression model.
Week 9	Implemented market-trend classification.
Week 10	Evaluated models and reviewed metric interpretation.
Week 11	Integrated ML service with backend marketplace workflow.
Week 12	Integrated seller-facing prediction views and history.
Week 13	Tested edge cases and model-unavailable behaviour.
Week 14	Reviewed limitations and real-data validation requirements.
Week 15	Prepared dissertation results, discussion and commercialization analysis.
Week 16	Final integration review, evidence planning and documentation.

Table 20 Contribution Timeline

Appendix H: Final Submission Checklist

- Candidate name and student ID verified: Thalagala T.A.D.V / IT22281678.
- SLIIT A4 margins applied.
- Times New Roman applied.
- Body text 12 pt and 1.5 line spacing applied.
- Chapter headings 14 pt bold uppercase.
- Figures and tables numbered and captioned.
- Member 04 ownership kept separate from Members 01–03.
- Synthetic-data limitation stated wherever model results are discussed.
- No fabricated screenshots included.

- References and in-text citations reviewed.
- Final DOCX and PDF generated and visually checked.

Appendix I: Extended Validation Notes

I.1 Classification balance

Final evaluation should inspect the distribution of Sold and Unsold classes before relying on accuracy. If the classes are imbalanced, precision, recall and F1 should be reported per class and macro-averaged where appropriate.

I.2 Temporal validation

Marketplace prediction should eventually use a chronological split so that the model is trained on earlier observations and evaluated on later observations. Random splits can allow future-like patterns to leak into the training set when temporal dependence is strong.

I.3 Sales-time censoring

A production dataset should distinguish completed sales from active listings that have not yet sold. Active listings do not necessarily have a final days-to-sell value and should not be incorrectly treated as zero or as an arbitrary upper bound.

I.4 Prediction calibration

Where probabilities are shown to users, calibration should be evaluated so that a stated 70% probability corresponds reasonably to an observed frequency over comparable cases.

I.5 Drift monitoring

Feature distributions, target rates and error measures should be monitored over time. Large changes can trigger investigation or retraining.

I.6 Human oversight

Sellers should remain able to review and modify listing decisions. AI predictions should not automatically reject listings or restrict seller access without a transparent human-review mechanism.